Executive Summary

A file system extraction of an Apple iPhone SE running iOS 13.4.1 was acquired, integrity-verified and examined. The device was identified by its serial number, ICCID, IMSI and registered phone number, and linked to an iCloud account (thisisdfir@gmail.com). The last iTunes/iCloud sync occurred on 16 April 2020. Messages, contacts, call history, Safari history and calendar artefacts were located and documented. Sixteen items of evidence (E01–E16) are listed in Section 5.

Scope and Method

The examination was based on the supplied archive ios_13_4_1.zip. The archive was hashed (MD5 and SHA-256), unzipped, and the extracted 13-4-1.tar was hashed again to confirm integrity. The directory structure was then reviewed and the key SQLite databases and property lists were opened and inspected.

Setup

```
┌──(kali㉿kali)-[~]
└─$ mkdir -p ~/SBT-DF205_iOScase/{originals,working,exports,screenshot}
```

Folder layout structured for further examination and Analysis

1. Evidence Preservation

Item	Detail
Archive	**ios_13_4_1.zip (9,629,286,492 bytes)**
Archive MD5	**4e094801ba1e3b938b2c0cc18d0e0d1d**
Archive SHA-256	**61854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c154551**
Extracted file	**13-4-1.tar (17,223,813,120 bytes)**

Tar MD5	**c2a733e6db7af9be6bd0437fd7c765f8**
Tar SHA-256	**991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb**

```
┌──(kali㉿kali)-[~/SBT-DF205_iOScase]
└─$ hashdeep ios_13_4_1.zip
%%%% HASHDEEP-1.0
%%%% size,md5,sha256,filename
## Invoked from: /home/kali/SBT-DF205_iOScase
## $ hashdeep ios_13_4_1.zip
##
9629286492,4e094801ba1e3b938b2c0cc18d0e0d1d,761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c154551,/home/kali/SBT-DF205_iOScase/ios_13_4_1.zip
```

Figure 1. (E01) zip file hashes verified and md5 and Sha256 values successfully retrieved

1.1file unzipping to get the extraction

```
┌──(kali㉿kali)-[~/SBT-DF205_iOScase]
└─$ unzip -q ios_13_4_1.zip

┌──(kali㉿kali)-[~/SBT-DF205_iOScase]
└─$ ls -l
total 9403628
drwxrwxr-x 2 kali kali       4096 Nov  4 18:32  exports
drwxr-xr-x 6 kali kali       4096 Apr 19  2020 'iOS 13.4.1 Extraction'
-rwxr----- 1 kali kali 9629286492 Nov  4 03:55  ios_13_4_1.zip
drwxrwxr-x 2 kali kali       4096 Nov  4 18:58  originals
drwxrwxr-x 2 kali kali       4096 Nov  4 18:32  screenshot
drwxrwxr-x 2 kali kali       4096 Nov  4 18:32  working
```

Figure 2. file quietly unzipped and listed with sizes and extraction

```
┌──(kali㉿kali)-[~/SBT-DF205_iOScase/iOS 13.4.1 Extraction/Extraction]
└─$ tar -xvf 13-4-1.tar

Volumes/JOSH/NoTar-13-4-1/
Volumes/JOSH/NoTar-13-4-1/.ba/
Volumes/JOSH/NoTar-13-4-1/.file
Volumes/JOSH/NoTar-13-4-1/.mb/
Volumes/JOSH/NoTar-13-4-1/.Trashes
```

```
┌──(kali㉿kali)-[~/SBT-DF205_iOScase/iOS 13.4.1 Extraction/Extraction]
└─$ hashdeep 13-4-1.tar
%%%% HASHDEEP-1.0
%%%% size,md5,sha256,filename
## Invoked from: /home/kali/SBT-DF205_iOScase/iOS 13.4.1 Extraction/Extraction
## $ hashdeep 13-4-1.tar
##
17223813120,c2a733e6db7af9be6bd0437fd7c765f8,991727d92145a08623e998c008cc2a05f086be722ee2945
815d0eb46a78f04fb,/home/kali/SBT-DF205_iOScase/iOS 13.4.1 Extraction/Extraction/13-4-1.tar
```

figure 3. tar file unzipped and hashed for integrity purpose

File name	13-4-1.tar
Size	17223813120
Md5	c2a733e6db7af9be6bd0437fd7c765f8
Sha256sum	991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb

1.2Device (ios) Root Level

```
┌──(kali㉿kali)-[~/…/iOS 13.4.1 Extraction/Extraction/Volumes/JOSH]
└─$ NoTar-13-4-1

┌──(kali㉿kali)-[~/…/Extraction/Volumes/JOSH/NoTar-13-4-1]
└─$ ls -l
total 40
drwxrwxr-x  2 kali kali 4096 Nov  5 15:23 Applications
drwxr-xr-x  2 kali kali 4096 Nov  5 15:23 bin
drwxrwxr-x  2 kali kali 4096 Apr  2  2020 cores
dr-xr-xr-x  2 kali kali 4096 Apr 16  2020 dev
drwxrwxr-x  2 kali kali 4096 Apr  2  2020 Developer
lrwxrwxrwx  1 kali kali   11 Apr  2  2020 etc → private/etc
drwxrwxr-x  2 kali kali 4096 Nov  5 15:24 Library
drwxr-xr-x  2 kali kali 4096 Nov  5 15:13 private
drwxr-xr-x  2 kali kali 4096 Nov  5 15:24 sbin
drwxr-xr-x  3 kali kali 4096 Nov  5 15:06 System
lrwxrwxrwx  1 kali kali   15 Apr  2  2020 tmp → private/var/tmp
drwxr-xr-x 10 kali kali 4096 Nov  5 15:06 usr
lrwxrwxrwx  1 kali kali   11 Apr  2  2020 var → private/var
```

Figure 4. (E02) ios device root level view (Comprising of directories for different functions)

1.3Vrgfb

2. iphone Device Information

2.1Software builds version

```
┌──(kali㉿kali)-[~/…/var/installd/Library/MobileInstallation]
└─$ plistutil -i  LastBuildInfo.plist
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
        <key>SystemImageID</key>
        <string>747CABD1-394F-4F90-92E9-CC6B5E1699DE</string>
        <key>Version</key>
        <string>Version</string>
        <key>ProductVersion</key>
        <string>13.4.1</string>
        <key>BuildID</key>
        <string>9695BE08-7547-11EA-B2DF-F0563A0B6C5E</string>
        <key>ProductBuildVersion</key>
        <string>17E262</string>
        <key>ProductCopyright</key>
        <string>1983-2020 Apple Inc.</string>
        <key>Build</key>
        <string>Build</string>
        <key>ProductName</key>
        <string>iPhone OS</string>
        <key>FullVersionString</key>
        <string>Version 13.4.1 (Build 17E262)</string>
</dict>
</plist>
```

Figure 5. (E03). The IOS device software build version is 13.4.1, product Name iphone OS and build number 17E262 as shown above and

2.2Phone number attached to the ios device

```
┌──(kali㉿kali)-[~/…/var/wireless/Library/Preferences]
└─$ plistutil -i com.apple.commcenter.plist
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
        <key>SIMPhoneNumber</key>
        <string>+19195794674</string>
        <key>PersonalitySlots</key>
        <array>
                <string>8901260971148676693</string>
        </array>
        <key>RoamingMigrationDone</key>
        <true/>
        <key>unique-sim-label-store</key>
        <dict>
                <key>0DC88727-B856-4F9C-B802-D838D799CA01</key>
                <dict>
                        <key>text</key>
                        <string></string>
                        <key>ts</key>
                        <integer>1584826745</integer>
```

Figure 6. (E04) Com.apple.commcenter.plist – Found at ~/private/var/wireless/Library/Preferences. The phone number registered on the SIM card inside this iPhone is +19195794674

2.3The device ICCID (Integrated Circuit Card Identifier)

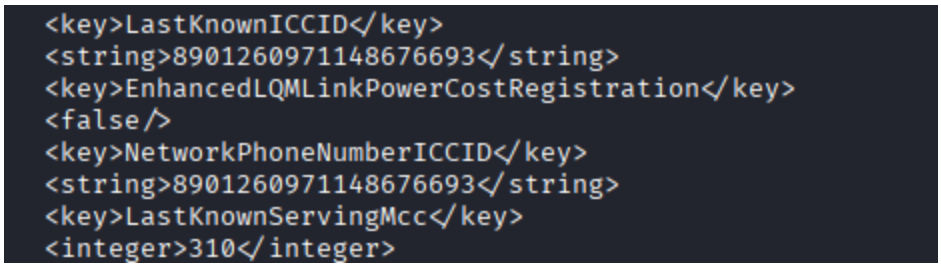


Figure 7. (E05) the card identifier was 8901260971148676693

2.4The Device IMSI

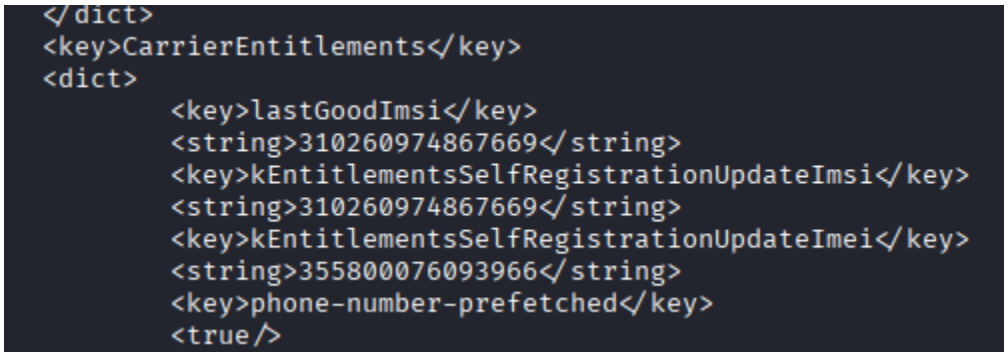


Figure 8. (E06) the device's International Mobile Subscriber Identity used to identify mobile network and users is 310260974867669 as shown.

2.5 The Device Serial Number

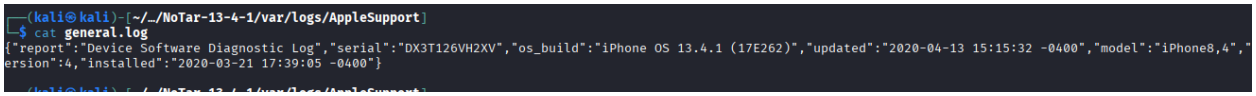


Figure 9. (E07) the unique piece identifier is DX3T126VH2XV

2.6Device's Model Number

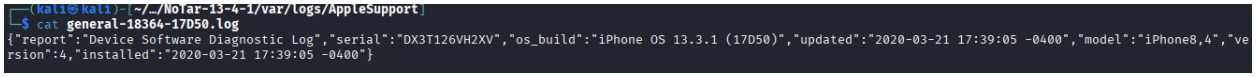


Figure 10. (E08) the generation name came as iphoen SE and model is iphone 8,4

3. Account Information (Device accounts and settings)

The iOS Account information is stored in a database file called Accounts3.sqlite.

3.1the icloud Account present

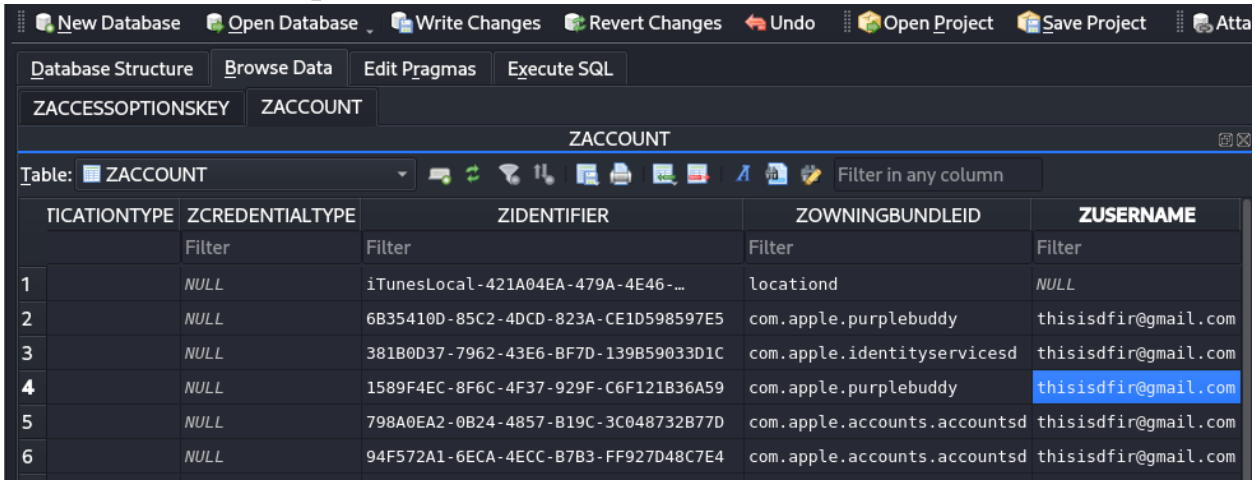
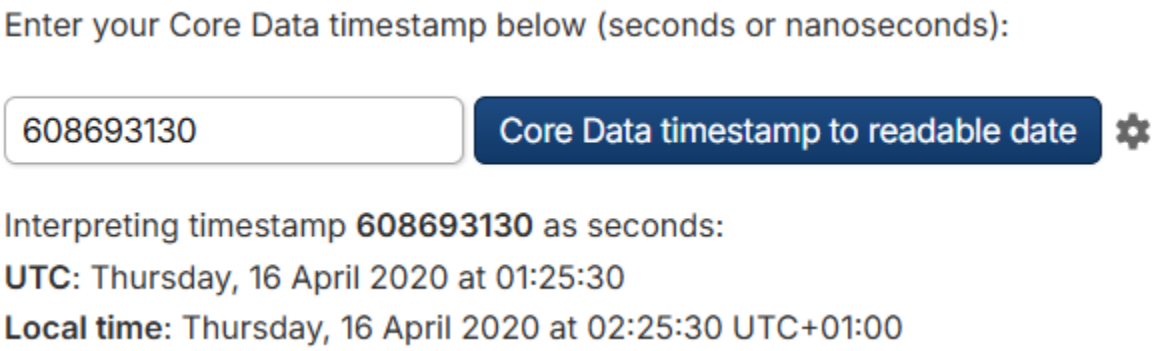


Figure 11. (E09) suucesfully retrieved the icloud inked to this ios device.

3.2Last iTunes & iCloud Backup date

```
┌──(kali㉿kali)-[~/…/var/mobile/Library/Preferences]
└─$ plistutil -i com.apple.mobile.ldbackup.plist
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
        <key>CloudBackupEnabled</key>
        <true/>
        <key>Version</key>
        <string>2.0</string>
        <key>LastiTunesBackupTZ</key>
        <string>EDT</string>
        <key>LastiTunesBackupDate</key>
        <integer>608744930</integer>
        <key>LastCloudBackupTZ</key>
        <string>EDT</string>
        <key>WillEncrypt</key>
        <true/>
        <key>RequiresEncryption</key>
        <integer>0</integer>
        <key>LastCloudBackupDate</key>
        <integer>608693130</integer>
</dict>
</plist>
```



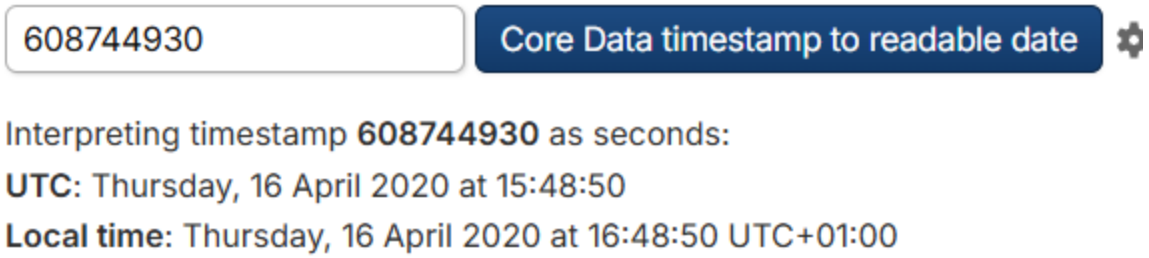
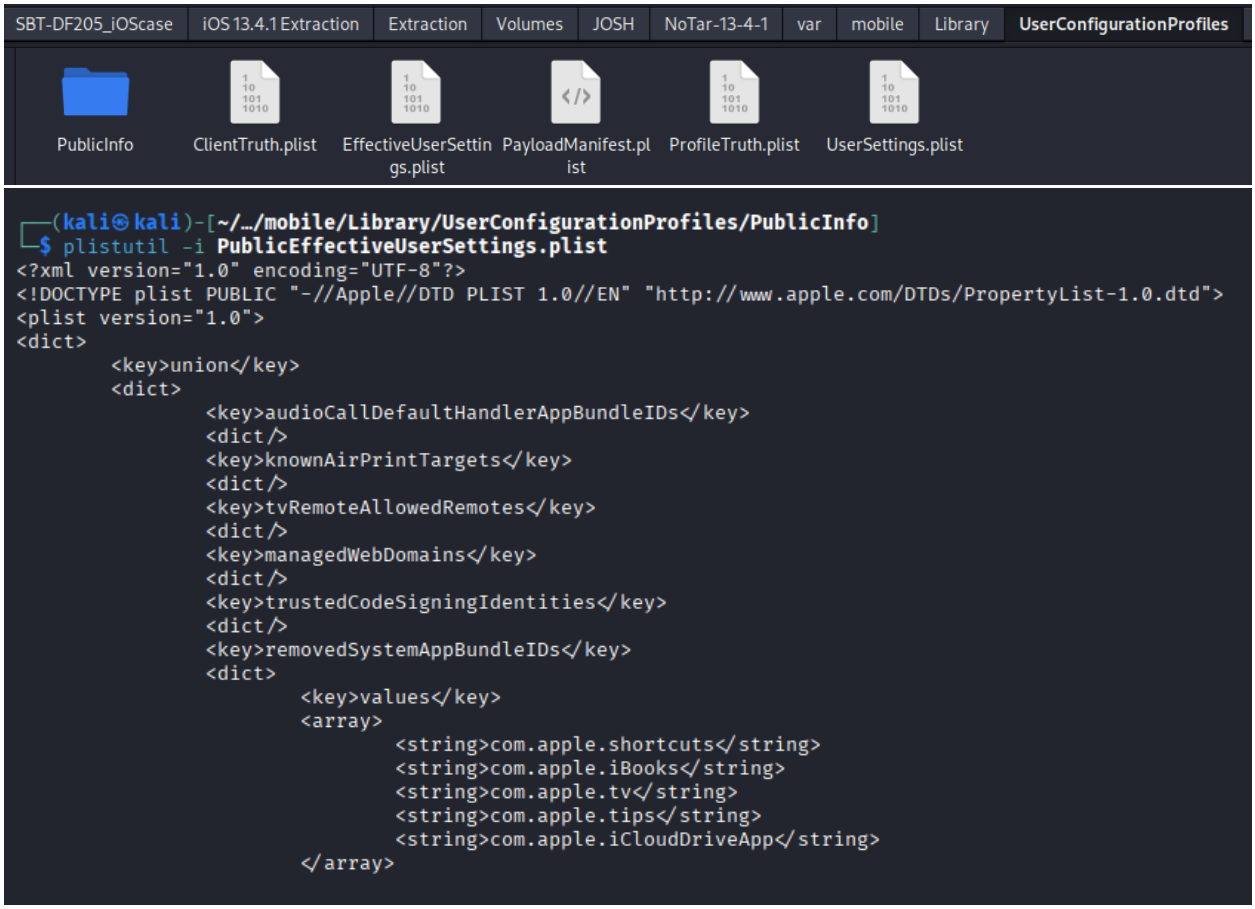


Figure 12. (E10). Data for the last time an iPhone was synced with iTunes or iCloud is 608693130 and 608744930 (Cocoa Core Date with timestamp January 1, 2001, GMT.), converted to human date and time using epochconverter.com/coredata to Thursday April 16 2020 01:25:30 and Thursday April, 16 2020 03:48:50 pm UTC

3.3Finding where settings are stored



```
        <key>value</key>
        <true/>
</dict>
<key>allowTV</key>
<dict>
        <key>value</key>
        <true/>
</dict>
<key>forceAutomaticDateAndTime</key>
<dict>
        <key>value</key>
        <false/>
</dict>
<key>allowDefinitionLookup</key>
<dict>
        <key>value</key>
```

Figure 13. (E11) the settings were found in this directory UserConfigurationProfiles with example of an XML on automatic time detection, showing false meaning the time is not set to be updated automatically.

4. Investigating Message (contacts and calls)

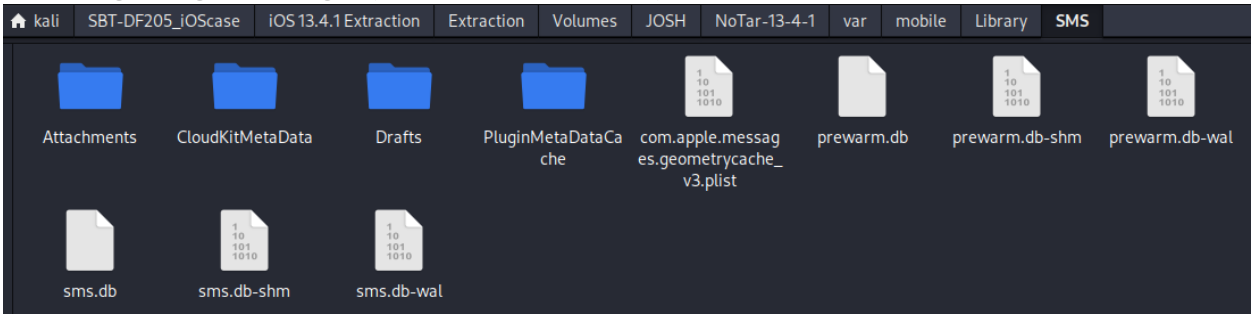


Figure 14. (E12) the SMS database with multiple directories with like attachments and Drafts (typed out but unsent) messages are also stored.

4.1Opening Sms Database

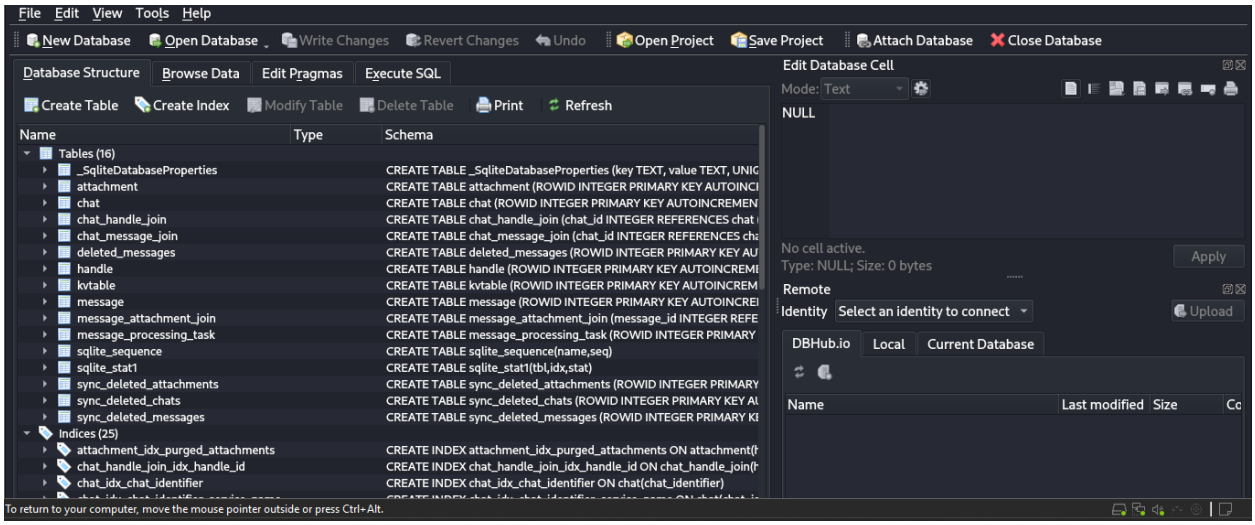
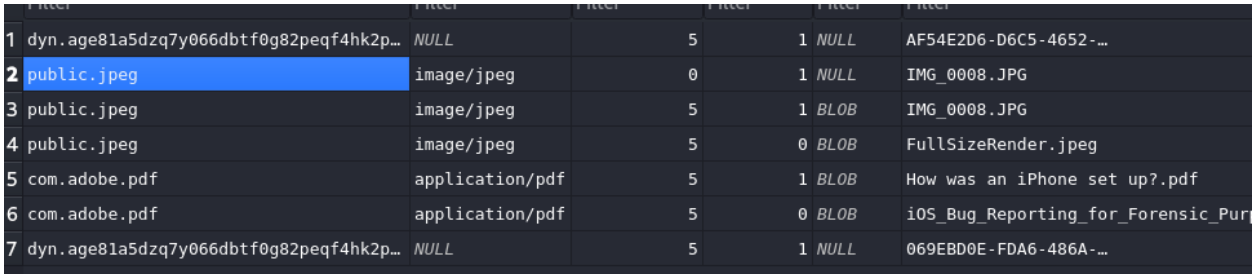


Figure 15. SMS database initial triage

4.2Investigating for attachment table



1	dyn.age81a5dzq7y066dbtf0g82peqf4hk2p…	*NULL*	5	1	*NULL*	AF54E2D6-D6C5-4652-…
2	public.jpeg	image/jpeg	0	1	*NULL*	IMG_0008.JPG
3	public.jpeg	image/jpeg	5	1	*BLOB*	IMG_0008.JPG
4	public.jpeg	image/jpeg	5	0	*BLOB*	FullSizeRender.jpeg
5	com.adobe.pdf	application/pdf	5	1	*BLOB*	How was an iPhone set up?.pdf
6	com.adobe.pdf	application/pdf	5	0	*BLOB*	iOS_Bug_Reporting_for_Forensic_Pur
7	dyn.age81a5dzq7y066dbtf0g82peqf4hk2p…	*NULL*	5	1	*NULL*	069EBD0E-FDA6-486A-…

Figure 16. (E13) attachment sms database with 7 transfer name attachments.

4.3Message Table database

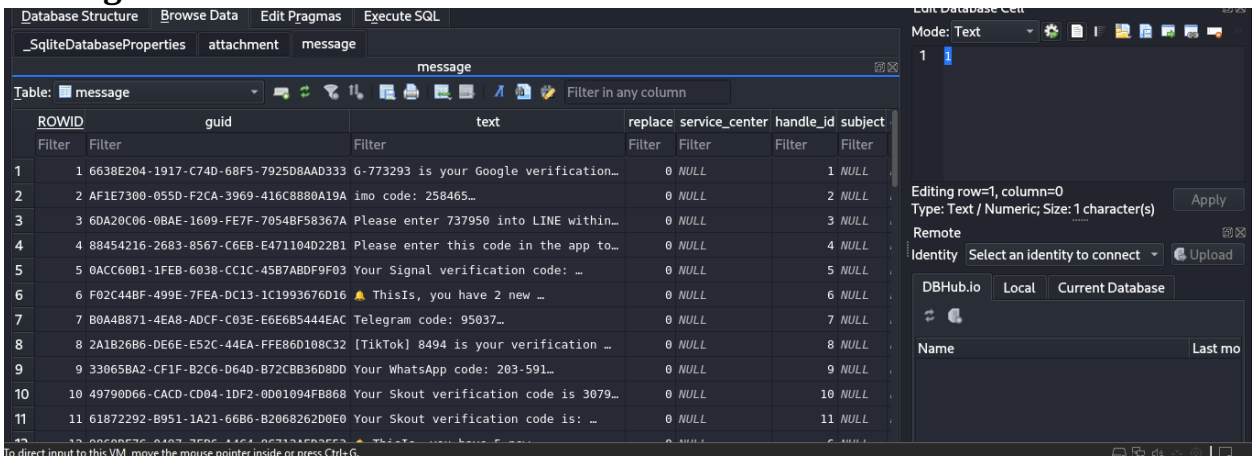


Figure 17 (E14) Message Table displayed with handle Id and 5 messages as requested and also extracted to a .csv file for evidence

4.4Investigating Contacts

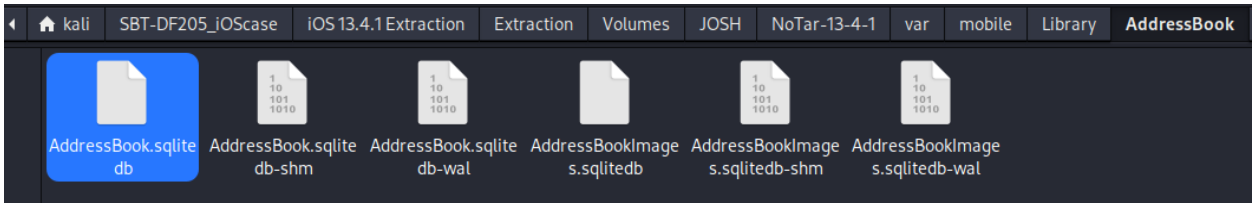


Figure 18. addressbook initial triage

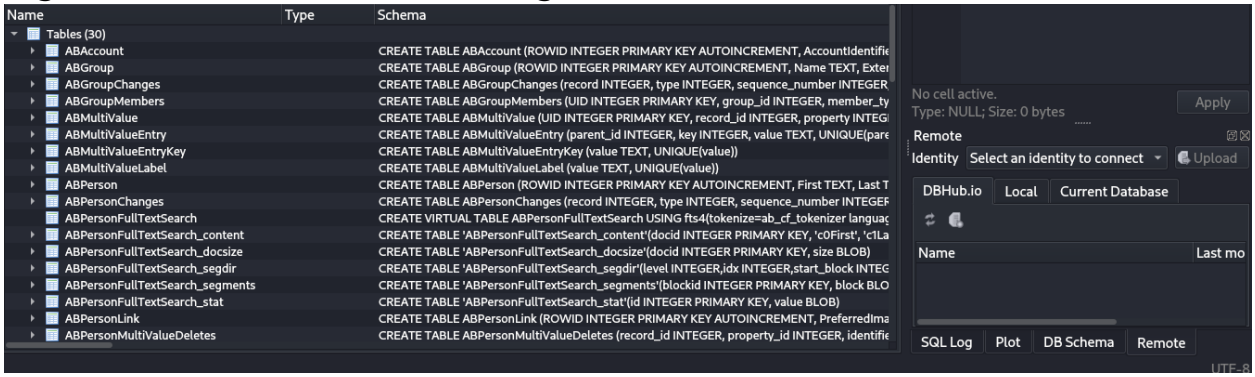
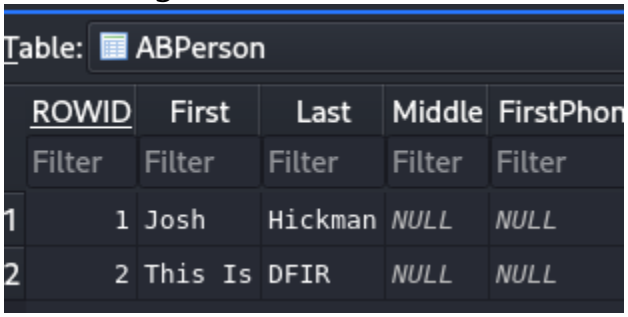


Figure 19. (E15) addressBook with multiple Tables that Stores the extended information from the ABPerson table like email address, phone numbers, website URL etc. The ABMultiValue tables uses a file named record_Id that is linked with the row_Id file of the ABPerson table.

Examining ABPerson



ABPerson with ROWID and their First and Last Name.

4.5Investigating Calls History

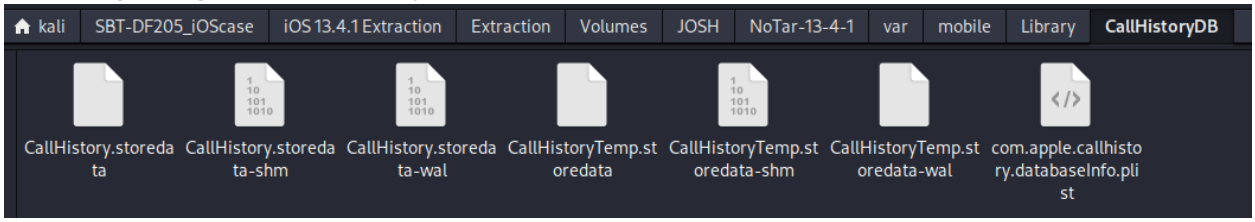


Figure 20. Call History database initial triage

ZCALLDBPROPERTIES | ZCALLRECORD

ZCALLRECORD

Table: ZCALLRECORD — Filter in any column

	TUS	ZDATE	ZDURATION	ZISO_COUNTRY_CODE	ZLOCATION	ZNAME	ZSERVICE_PROVIDER
		Filter	Filter	Filter	Filter	Filter	Filter
1	4	606686572.413675	0.0	us	San Jose, CA	NULL	com.apple.Telephony
2	4	606764238.386421	0.0	us	San Jose, CA	NULL	com.apple.Telephony
3	4	606937905.317984	0.0	us	San Jose, CA	NULL	com.apple.Telephony
4	4	607019136.714309	0.0	us	San Jose, CA	NULL	com.apple.Telephony
5	4	607031703.669631	0.0	us	San Jose, CA	NULL	com.apple.Telephony
6	4	607464398.417464	0.0	us	San Jose, CA	NULL	com.apple.Telephony
7	4	607623054.597476	0.0	us	San Jose, CA	NULL	com.apple.Telephony
8	4	607812138.039618	23.08800303936	us	Fuquay-Varina, NC	NULL	com.apple.Telephony
9	4	607898073.986463	0.0	us	San Jose, CA	NULL	com.apple.Telephony

1 - 9 of 37 — Go to: 1

Main table where call history and metadata are stored, in the ZCALLRECORD where date, duration of the call, country code and locations were recorded.

5. Safari and calendar Investigation

5.1Finding Evidence generated by Safari

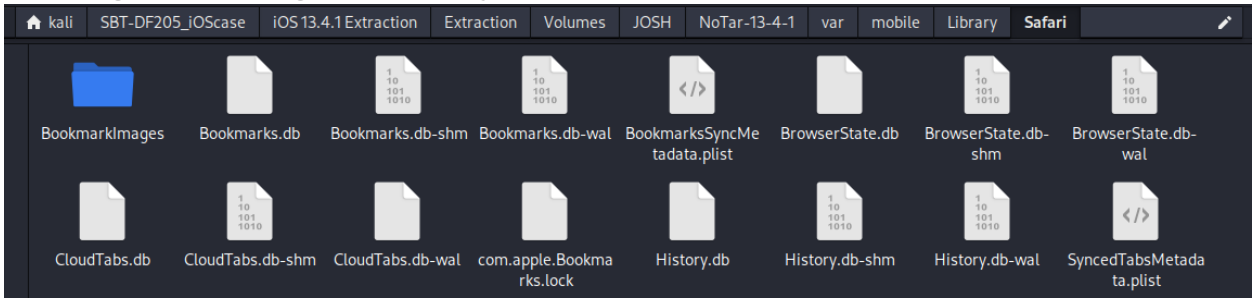
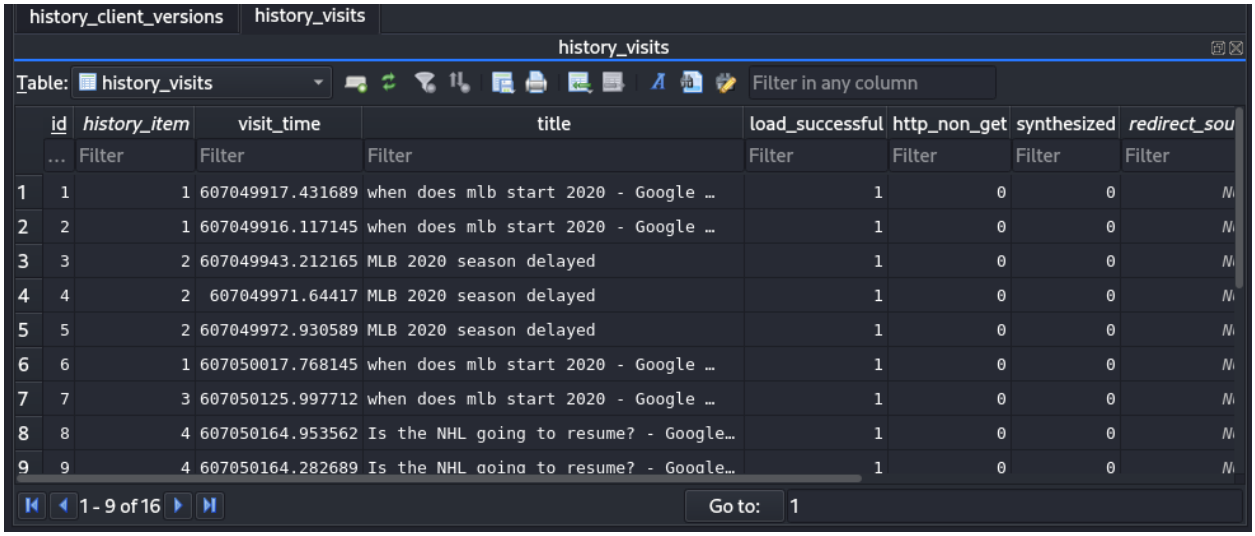


Figure 21. Safari Initial Triage

5.2History Visit



	id	history_item	visit_time	title	load_successful	http_non_get	synthesized	redirect_sou
	...	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	1	607049917.431689	when does mlb start 2020 - Google …	1	0	0	N
2	2	1	607049916.117145	when does mlb start 2020 - Google …	1	0	0	N
3	3	2	607049943.212165	MLB 2020 season delayed	1	0	0	N
4	4	2	607049971.64417	MLB 2020 season delayed	1	0	0	N
5	5	2	607049972.930589	MLB 2020 season delayed	1	0	0	N
6	6	1	607050017.768145	when does mlb start 2020 - Google …	1	0	0	N
7	7	3	607050125.997712	when does mlb start 2020 - Google …	1	0	0	N
8	8	4	607050164.953562	Is the NHL going to resume? - Google…	1	0	0	N
9	9	4	607050164.282689	Is the NHL going to resume? - Google…	1	0	0	N

Figure 22. (E16) history visits table with visited timestamp where Google Searches, searches are marked with –Google Search

5.3Inspecting Calender

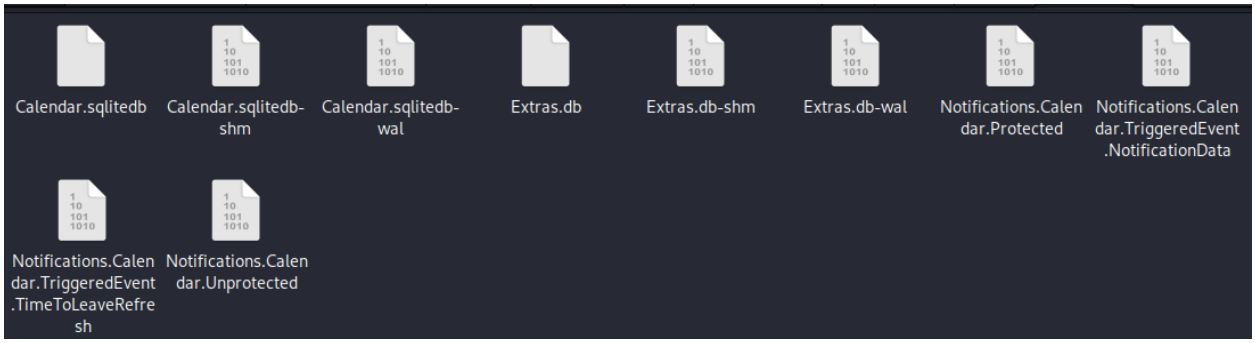


Figure 23. Initial Tirae of the Calender Database

Finding the location of the evidence

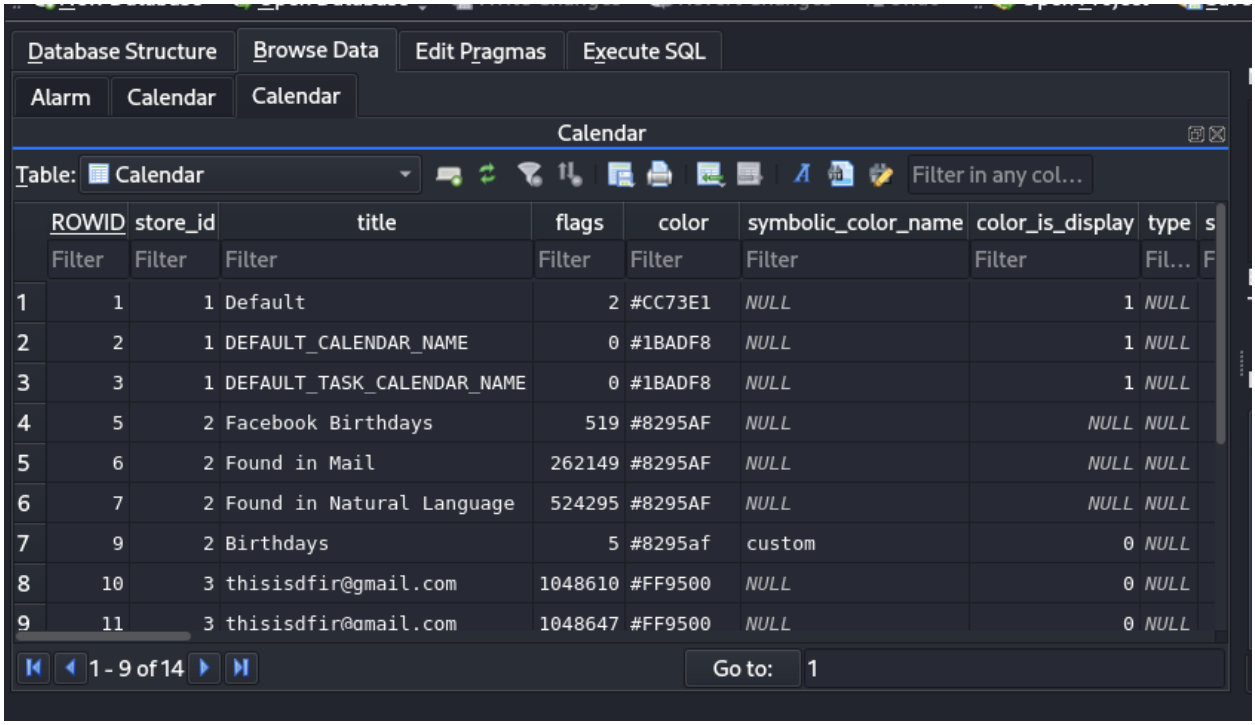


Figure 24. (E17) Calender table with titles of different events, flags and color display

5.4Examining Extra Database

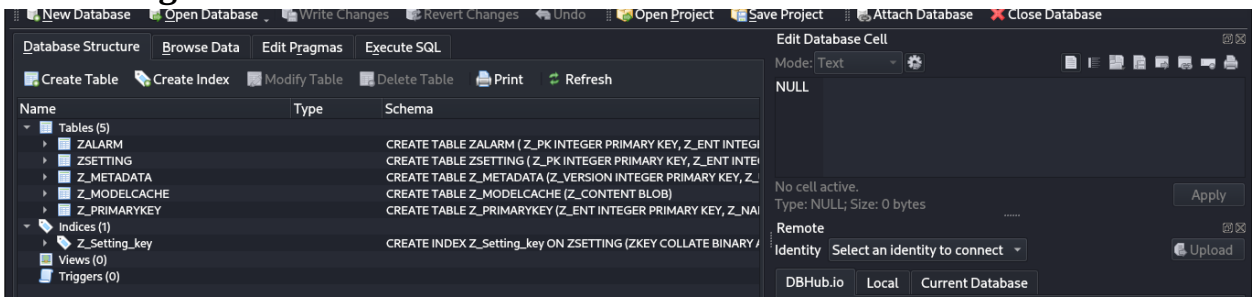


Figure 25. (E18) the extra database holds extra Data for the main Calendar database

Evidence Register (E01–E16)

ID	Evidence	Source / Location	Finding
E01	Evidence archive hashes	ios_13_4_1.zip	MD5 and SHA-256 values successfully retrieved; integrity verified

ID	Evidence	Source / Location	Finding
E02	Device root level	Extracted file system	Root directories for the different device functions
E03	Software build version	iOS version information	iOS 13.4.1, build 17E262, product name iPhone OS
E04	Registered phone number	com.apple.commcenter.plist (~/private/var/wireless/Library/Preferences)	+19195794674
E05	ICCID	SIM information	8901260971148676693
E06	IMSI	SIM information	310260974867669
E07	Serial number	Device information	DX3T126VH2XV
E08	Model	Device information	iPhone SE, model iPhone8,4
E09	iCloud account	Accounts3.sqlite (ZACCOUNT table)	iCloud account thisisdfir@gmail.com linked to the device
E10	Last iTunes/iCloud backup	Backup timestamps (Cocoa Core Data)	608693130 and 608744930, converted to Thu 16 Apr 2020, 01:25:30 and 03:48:50 pm UTC
E11	Device settings	UserConfiguration Profiles (XML)	Automatic time detection set to false
E12	SMS database	SMS directory	Contains attachments and drafts (typed but unsent messages)
E13	SMS attachment table	SMS database	7 transfer name attachments
E14	SMS message table	SMS database	Messages with handle ID, 5 messages; extracted to a .csv file
E15	Address book	AddressBook database	ABPerson and ABMultiValue tables hold emails, phone numbers

ID	Evidence	Source / Location	Finding
			and URLs, linked by record_id and row_id
E16	Safari history visits	Safari history database	Visit timestamps recorded; Google searches marked as Google Search